Homework 02: Solutions
REAL VARIABLES I
Fall 2026

Solutions for Homework 02.

1. (Images of functions with bounded derivative)

1.a

Minimal solution. Since $f(\mathbb{R}) = \bigcup_{N\in\mathbb{N}} f([-N,N])$, by countable subadditivity of $\mathcal{L}_2^*$ it suffices to show that $\mathcal{L}_2^*(f([-N,N])) = 0$ for every $N \in \mathbb{N}$. Fix N.
Write $f = (f_1, f_2)$. Fix n and split $[-N,N]$ into n intervals of length $2N/n$:

$$[-N,N] = \bigcup_{k=0}^{n-1} [t_k, t_{k+1}],$$

where $t_k = -N + 2Nk/n$ for $k = 0, \dots, n$. Since $f([-N,N]) = \bigcup_{k=0}^{n-1} f([t_k, t_{k+1}])$, subadditivity of $\mathcal{L}_2^*$ gives

$$\mathcal{L}_2^*(f([-N,N])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(f([t_k, t_{k+1}])).$$

By the mean value theorem, $|f_i(s) - f_i(t)| \le \|f'\|_{\sup}|s-t|$ for $i = 1, 2$. Therefore $f([t_k, t_{k+1}])$ lies in the closed square $Q_{n,k}$ of side $4N\|f'\|_{\sup}/n$ centered at $f(t_k)$. By monotonicity of $\mathcal{L}_2^*$ and $\mathcal{L}_2^*$(closed rectangle) $= \mathrm{Vol}_2$,

$$\mathcal{L}_2^*(f([t_k, t_{k+1}])) \le \mathcal{L}_2^*(Q_{n,k}) = \mathrm{Vol}_2(Q_{n,k}) = \Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2.$$

Hence

$$\mathcal{L}_2^*(f([-N,N])) \le n\Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2 = \frac{16N^2\|f'\|_{\sup}^2}{n}.$$

Since n is arbitrary, the claim follows by taking $n \to \infty$.

Solution. Write $f = (f_1, f_2)$. The derivative of f is bounded, so $\|f'\|_{\sup} < \infty$. We show that $\mathcal{L}_2^*(f(\mathbb{R})) = 0$. We have $f(\mathbb{R}) = \bigcup_{N\in\mathbb{N}} f([-N,N])$. By countable subadditivity of $\mathcal{L}_2^*$, it suffices to show that $\mathcal{L}_2^*(f([-N,N])) = 0$ for every $N \in \mathbb{N}$. Fix N.
Step 1: $|f_i(s) - f_i(t)| \le \|f'\|_{\sup}|s-t|$ for all $s,t \in \mathbb{R}$ and $i = 1,2$. For $s = t$ both sides vanish. For $s \ne t$ the mean value theorem gives ξ between s and t with $|f_i(s) - f_i(t)| = |f_i'(\xi)||s-t|$. Moreover $|f_i'(\xi)| \le \|f'(\xi)\| \le \|f'\|_{\sup}$.
Step 2: $f([-N,N])$ is the union of n sets, each inside a closed square. Fix $n \ge 1$. Let $t_k := -N + 2Nk/n$ for $k = 0, \dots, n$. Then $[-N,N] = \bigcup_{k=0}^{n-1}[t_k, t_{k+1}]$, and hence $f([-N,N]) = \bigcup_{k=0}^{n-1} f([t_k, t_{k+1}])$. Let $Q_{n,k}$ be the closed square with center $f(t_k)$ and side $4N\|f'\|_{\sup}/n$. Take $t \in [t_k, t_{k+1}]$. Then $|t - t_k| \le 2N/n$. By Step 1, $|f_i(t) - f_i(t_k)| \le 2N\|f'\|_{\sup}/n$ for $i = 1, 2$. Therefore $f(t) \in Q_{n,k}$, and so $f([t_k, t_{k+1}]) \subseteq Q_{n,k}$.
Step 3: $\mathcal{L}_2^*(Q_{n,k}) \le \mathrm{Vol}_2(Q_{n,k})$. Let $\delta > 0$. The square $Q_{n,k}$ is contained in the open square with the same center and side $4N\|f'\|_{\sup}/n + 2\delta$. Since $\mathcal{L}_2^*$ of an open rectangle is its volume, monotonicity gives $\mathcal{L}_2^*(Q_{n,k}) \le (4N\|f'\|_{\sup}/n + 2\delta)^2$. This holds for every $\delta > 0$. Letting $\delta \to 0$ gives $\mathcal{L}_2^*(Q_{n,k}) \le (4N\|f'\|_{\sup}/n)^2 = \mathrm{Vol}_2(Q_{n,k})$.
Step 4: conclusion. By subadditivity and then monotonicity of $\mathcal{L}_2^*$, Steps 2 and 3 give

$$\mathcal{L}_2^*(f([-N,N])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(f([t_k, t_{k+1}])) \le \sum_{k=0}^{n-1} \mathcal{L}_2^*(Q_{n,k}) \le n\Big(\frac{4N\|f'\|_{\sup}}{n}\Big)^2 = \frac{16N^2\|f'\|_{\sup}^2}{n}.$$

The left side does not depend on n, and $\|f'\|_{\sup}$ and N do not depend on n. Since n is arbitrary, the claim follows by taking $n \to \infty$.

2. (Equivalent interval definitions of Lebesgue outer measure)

2.a

Solution. Let $\mu_o^* = \mathcal{L}_1^*$, μ_c^*, μ_d^* be the outer measures generated by length on bounded open, bounded closed, and dyadic intervals (in each family $\emptyset$ is admitted with cost 0). We show $\mu_o^* = \mu_c^* = \mu_d^*$. Here μ_o^* is $\mathcal{L}_1^*$ by definition.

Lemma. Let $\mathcal{A}, \mathcal{F} \subseteq 2^X$ contain $\emptyset$, with costs $c_{\mathcal{A}}, c_{\mathcal{F}}$ vanishing at $\emptyset$, and let $\mu_{\mathcal{A}}^*$, $\mu_{\mathcal{F}}^*$ be the generated outer measures. Suppose that for every $A \in \mathcal{A}$ and every $\delta > 0$ there is a countable cover $(F_j)_j \subseteq \mathcal{F}$ of A with $\sum_j c_{\mathcal{F}}(F_j) \le c_{\mathcal{A}}(A) + \delta$. Then $\mu_{\mathcal{F}}^* \le \mu_{\mathcal{A}}^*$.

Proof. Let $E \subseteq X$. If $\mu_{\mathcal{A}}^*(E) = +\infty$, the inequality holds because its right side is $+\infty$. Otherwise $\mu_{\mathcal{A}}^*(E) < \infty$; fix $\epsilon > 0$. It suffices to construct a countable cover of E by members of $\mathcal{F}$ of total cost at most $\mu_{\mathcal{A}}^*(E) + 2\epsilon$. Indeed, this gives $\mu_{\mathcal{F}}^*(E) \le \mu_{\mathcal{A}}^*(E) + 2\epsilon$. Since $\epsilon > 0$ is arbitrary, $\mu_{\mathcal{F}}^*(E) \le \mu_{\mathcal{A}}^*(E)$.

Take a cover $(A_i)_{i\in\mathbb{N}} \subseteq \mathcal{A}$ of E with $\sum_{i\in\mathbb{N}} c_{\mathcal{A}}(A_i) \le \mu_{\mathcal{A}}^*(E) + \epsilon$. For each i, take a cover $(F_{i,j})_{j\in\mathbb{N}} \subseteq \mathcal{F}$ of A_i with $\sum_{j\in\mathbb{N}} c_{\mathcal{F}}(F_{i,j}) \le c_{\mathcal{A}}(A_i) + \epsilon 2^{-i-1}$. This is possible by the hypothesis with $\delta = \epsilon 2^{-i-1}$. A finite cover is padded with $\emptyset \in \mathcal{F}$, which costs 0. The family $(F_{i,j})_{(i,j)\in\mathbb{N}^2}$ lies in $\mathcal{F}$ and is countable. Enumerate $\mathbb{N}^2$ by a bijection with $\mathbb{N}$. The total cost is a sum of nonnegative terms, so it does not depend on the enumeration. The family covers E, because the A_i cover E and the $F_{i,j}$ cover A_i. Its total cost is at most $\sum_{i\in\mathbb{N}}(c_{\mathcal{A}}(A_i) + \epsilon 2^{-i-1})$. This is at most $\mu_{\mathcal{A}}^*(E) + 2\epsilon$, since $\sum_{i\in\mathbb{N}} c_{\mathcal{A}}(A_i) \le \mu_{\mathcal{A}}^*(E) + \epsilon$ and $\sum_{i\in\mathbb{N}} \epsilon 2^{-i-1} = \epsilon$.

By the Lemma, each of (i)–(iv) follows from its hypothesis. For every interval A of the first family and every $\delta > 0$ we exhibit a countable cover of A by intervals of the second family of total length at most the length of A plus δ. For $A = \emptyset$ the cover $(\emptyset)$ has total cost 0. Hence we may take $A \neq \emptyset$. In each item, $\mathcal{A}$ is the family generating the outer measure on the right of the inequality and $\mathcal{F}$ the family generating the one on the left. The four inequalities give $\mu_c^* \le \mu_o^* \le \mu_c^*$ and $\mu_d^* \le \mu_o^* \le \mu_d^*$. Hence $\mu_o^* = \mu_c^* = \mu_d^*$.

(i) $\mu_c^* \le \mu_o^*$**.** The open interval (a, b) is covered by the closed interval $[a, b]$ of the same length.

(ii) $\mu_o^* \le \mu_c^*$**.** A closed interval $[a, b]$ is covered by $(a - \delta/2, b + \delta/2)$, of length $b - a + \delta$.

(iii) $\mu_o^* \le \mu_d^*$**.** A dyadic interval $[k2^{-m}, (k + 1)2^{-m})$ is covered by $(k2^{-m} - \delta/2, (k + 1)2^{-m} + \delta/2)$, of length $2^{-m} + \delta$.

(iv) $\mu_d^* \le \mu_o^*$**.** Let $A = (a, b)$ with $a < b$ and let $\delta > 0$. Choose $m \in \mathbb{N}$ with $2 \cdot 2^{-m} \le \delta$. Then m depends only on δ, not on a, b. The dyadic intervals $[k2^{-m}, (k + 1)2^{-m})$, $k \in \mathbb{Z}$, partition $\mathbb{R}$. Therefore those meeting (a, b) cover (a, b). Such an interval meets (a, b) exactly when $a2^m - 1 < k < b2^m$. This is an open interval of length $(b - a)2^m + 1$. Hence it contains at most $(b - a)2^m + 2$ integers. Their total length is therefore at most $b - a + 2 \cdot 2^{-m} \le b - a + \delta$.

3. (Finite interval covers and outer Jordan content)

3.a

Minimal solution. For every $\delta > 0$, $(0, 1) \subseteq [0, 1] \subseteq (-\delta, 1+\delta)$. By monotonicity and $\mathcal{L}_1^*$(open interval) = length,

$$1 = \mathcal{L}_1^*((0, 1)) \le \mathcal{L}_1^*([0, 1]) \le \mathcal{L}_1^*((-\delta, 1 + \delta)) = 1 + 2\delta.$$

Letting $\delta \to 0$ gives $\mathcal{L}_1^*([0, 1]) = 1$. By monotonicity, $\mathcal{L}_1^*([0, 1] \setminus \mathbb{Q}) \le \mathcal{L}_1^*([0, 1]) = 1$. Since $[0, 1] \cap \mathbb{Q}$ is countable, $\mathcal{L}_1^*([0, 1] \cap \mathbb{Q}) = 0$ by the nullity of countable sets. By subadditivity,

$$1 = \mathcal{L}_1^*([0, 1]) \le \mathcal{L}_1^*([0, 1] \setminus \mathbb{Q}) + \mathcal{L}_1^*([0, 1] \cap \mathbb{Q}) = \mathcal{L}_1^*([0, 1] \setminus \mathbb{Q}).$$

Solution. We show that $\mathcal{L}_1^*([0, 1] \setminus \mathbb{Q}) = 1$.

Step 1: $\mathcal{L}_1^*([0,1]) = 1$**.** Fix $\delta > 0$; we show $1 \le \mathcal{L}_1^*([0,1]) \le 1 + 2\delta$. We have $(0,1) \subseteq [0,1] \subseteq (-\delta, 1+\delta)$. Since $\mathcal{L}_1^*$ of an open interval is its length and $\mathcal{L}_1^*$ is monotone, $1 \le \mathcal{L}_1^*([0,1]) \le 1 + 2\delta$. This holds for every $\delta > 0$, so letting $\delta \to 0$ gives the claim.
Step 2: conclusion. We show that $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \le 1$ and $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \ge 1$. The set $[0,1] \cap \mathbb{Q}$ is countable, hence $\mathcal{L}_1^*$-null. Monotonicity gives $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) \le \mathcal{L}_1^*([0,1]) = 1$. Subadditivity gives

$$1 = \mathcal{L}_1^*([0,1]) \le \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) + \mathcal{L}_1^*([0,1] \cap \mathbb{Q}) = \mathcal{L}_1^*([0,1] \setminus \mathbb{Q}).$$

3.b

Solution. We show that $J(E) = J(\bar{E})$ for every $E \subseteq \mathbb{R}$ (with $\inf \emptyset = +\infty$), and that $E = [0,1] \cap \mathbb{Q}$ satisfies $J(E) = 1$ and $\mathcal{L}_1^*(E) = 0$.
Step 1: $J(E) = J(\bar{E})$**.** A finite union U of closed intervals is closed. Hence $E \subseteq U$ implies $\bar{E} \subseteq U$. Conversely, $\bar{E} \subseteq U$ implies $E \subseteq U$. Therefore a finite collection of closed intervals has union containing E if and only if its union contains $\bar{E}$. Hence, according to the definition of J as an infimum over such collections, it holds that $J(E) = J(\bar{E})$.
Step 2: if $[c,d] \subseteq \bigcup_{n<N} [a_n, b_n]$**, then** $d - c \le \sum_{n<N} (b_n - a_n)$**.** If $c = d$, the claim holds because the right side is nonnegative. Let $c < d$ and fix $\delta > 0$. It suffices to show that $d - c \le \sum_{n<N} (b_n - a_n) + 2N\delta$ for every $\delta > 0$. The intervals, hence N, are fixed before δ. Therefore letting $\delta \to 0$ gives the claim. We have $(c,d) \subseteq [c,d] \subseteq \bigcup_{n<N} (a_n - \delta, b_n + \delta)$. Hence, by monotonicity, subadditivity, and $\mathcal{L}_1^*(\text{open interval}) = \text{length}$,

$$d - c \le \sum_{n<N} (b_n - a_n + 2\delta) = \sum_{n<N} (b_n - a_n) + 2N\delta.$$

Step 3: $J([0,1]) = 1$**.** The cover $\{[0,1]\}$ gives $J([0,1]) \le 1$, and Step 2 gives $J([0,1]) \ge 1$.
Step 4: the example. Let $E = [0,1] \cap \mathbb{Q}$. It is countable, so $\mathcal{L}_1^*(E) = 0$. Since $\mathbb{Q}$ is dense and $[0,1]$ is closed, $\bar{E} = [0,1]$. By Steps 1 and 3, $J(E) = J([0,1]) = 1 \ne 0 = \mathcal{L}_1^*(E)$.

Discussion. Finite covers see only the closure of E. Hence they cannot detect that a dense countable set is negligible. Countable covers can, by shrinking the j-th interval geometrically.